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In [33]: import os
from google import genai
from google.genai import types
from google.colab import userdata
from pydantic import BaseModel, Field
from typing import List, Literal

api_key = userdata.get('GEN_AI')
client = genai.Client(api_key=api_key)

class Transaction(BaseModel):
    date: str = Field(description="The date of the transaction (e.g., '3 November') or empty string if not mentioned.")
    merchant: str = Field(description="The name of the merchant, service provider, or location.")
    amount: float = Field(description="Numeric value of the transaction. Do not include currency symbols.")
    payment_method: str = Field(description="Payment method used. Use 'Unknown' if not mentioned.")
    category: Literal[
        "Accommodation", "Transportation", "Food", "Sightseeing",
        "Shopping", "Entertainment", "Adventure Activities",
        "Local Travel", "Fuel", "Others"
    ] = Field(description="The exact category classification based on guidelines.")

class TransactionList(BaseModel):
    transactions: List[Transaction]

passage = """
Ramesh planned a five-day vacation to Ooty during the month of November. On 3 November, he booked train tickets through IRCTC costing ₹2,450 using Net Banking. After reaching Ooty, he hired a taxi from the railway station to his hotel and paid ₹850 through UPI.

He checked into Hill View Resort, where the accommodation charges were ₹9,800 for two nights. The payment was made using his Credit Card. During the same evening, he had dinner at A2B Restaurant and spent ₹760 using Debit Card.

The following morning, he visited the Government Botanical Garden and purchased entry tickets worth ₹150. The payment method was not mentioned. Later, he booked a guided sightseeing tour of Doddabetta Peak costing ₹1,250 using UPI.

During the afternoon, Ramesh purchased homemade chocolates from Moddy's Chocolates worth ₹980 using Cash. He also bought woollen sweaters from a local textile showroom for ₹3,850. The payment method was not specified.

On 5 November, he rented a bicycle for local sightseeing and paid ₹450 through UPI. He later enjoyed a boating activity at Ooty Lake, paying ₹300 for the tickets. In the evening, he had lunch at Earl's Secret

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Restaurant, where the bill amounted to ₹1,180, paid using his Credit Card.

The next day, he booked a jeep safari to Avalanche Forest costing ₹2,600 through Net Banking. During the journey, he refuelled his rental vehicle at Indian Oil for ₹1,750 using Debit Card.

Before returning home, he purchased tea powder from the Government Tea Factory worth ₹1,450 using UPI and bought souvenirs for his family costing ₹2,250. The payment method for the souvenirs was not mentioned. Finally, he booked a return bus ticket through RedBus for ₹1,380 using Credit Card.

"""

prompt = f"""

Analyze the travel passage below and extract all financial transactions into a structured JSON array format.

Strict rules to follow:

- If a payment method is not mentioned, return "Unknown".
- 'amount' should contain only numeric values without the currency symbol (₹).
- Classify every transaction into one of these strict categories: Accommodation, Transportation, Food, Sightseeing, Shopping, Entertainment, Adventure Activities, Local Travel, Fuel, Others.

Passage to analyze:

{passage}

"""

print("Analyzing travel text using Gemini 2.5 Flash...")

```
response = client.models.generate_content(
    model='gemini-3.5-flash',
    contents=prompt,
    config=types.GenerateContentConfig(
        response_mime_type="application/json",
        response_schema=TransactionList,
        temperature=0.0
    ),
)
```

import json

parsed_json = json.loads(response.text)

print(json.dumps(parsed_json["transactions"], indent=2))

Analyzing travel text using Gemini 2.5 Flash...

```
[
  {
    "date": "3 November",
    "merchant": "IRCTC",
    "amount": 2450,
    "payment_method": "Net Banking",
    "category": "Transportation"
  },
  {
    "date": "3 November",
```

```
"merchant": "Taxi",
"amount": 850,
"payment_method": "UPI",
"category": "Local Travel"
},
{
  "date": "3 November",
  "merchant": "Hill View Resort",
  "amount": 9800,
  "payment_method": "Credit Card",
  "category": "Accommodation"
},
{
  "date": "3 November",
  "merchant": "A2B Restaurant",
  "amount": 760,
  "payment_method": "Debit Card",
  "category": "Food"
},
{
  "date": "4 November",
  "merchant": "Government Botanical Garden",
  "amount": 150,
  "payment_method": "Unknown",
  "category": "Sightseeing"
},
{
  "date": "4 November",
  "merchant": "Doddabetta Peak",
  "amount": 1250,
  "payment_method": "UPI",
  "category": "Sightseeing"
},
{
  "date": "4 November",
  "merchant": "Moddy's Chocolates",
  "amount": 980,
  "payment_method": "Cash",
  "category": "Shopping"
},
{
  "date": "4 November",
  "merchant": "Local Textile Showroom",
  "amount": 3850,
  "payment_method": "Unknown",
  "category": "Shopping"
},
{
  "date": "5 November",
  "merchant": "Bicycle Rental",
  "amount": 450,
  "payment_method": "UPI",
  "category": "Local Travel"
},
{
  "date": "5 November",
  "merchant": "Ooty Lake",
```

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"amount": 300,
"payment_method": "Unknown",
"category": "Entertainment"
},
{
  "date": "5 November",
  "merchant": "Earl's Secret Restaurant",
  "amount": 1180,
  "payment_method": "Credit Card",
  "category": "Food"
},
{
  "date": "6 November",
  "merchant": "Avalanche Forest",
  "amount": 2600,
  "payment_method": "Net Banking",
  "category": "Adventure Activities"
},
{
  "date": "6 November",
  "merchant": "Indian Oil",
  "amount": 1750,
  "payment_method": "Debit Card",
  "category": "Fuel"
},
{
  "date": "",
  "merchant": "Government Tea Factory",
  "amount": 1450,
  "payment_method": "UPI",
  "category": "Shopping"
},
{
  "date": "",
  "merchant": "Souvenirs",
  "amount": 2250,
  "payment_method": "Unknown",
  "category": "Shopping"
},
{
  "date": "",
  "merchant": "RedBus",
  "amount": 1380,
  "payment_method": "Credit Card",
  "category": "Transportation"
}
]
```